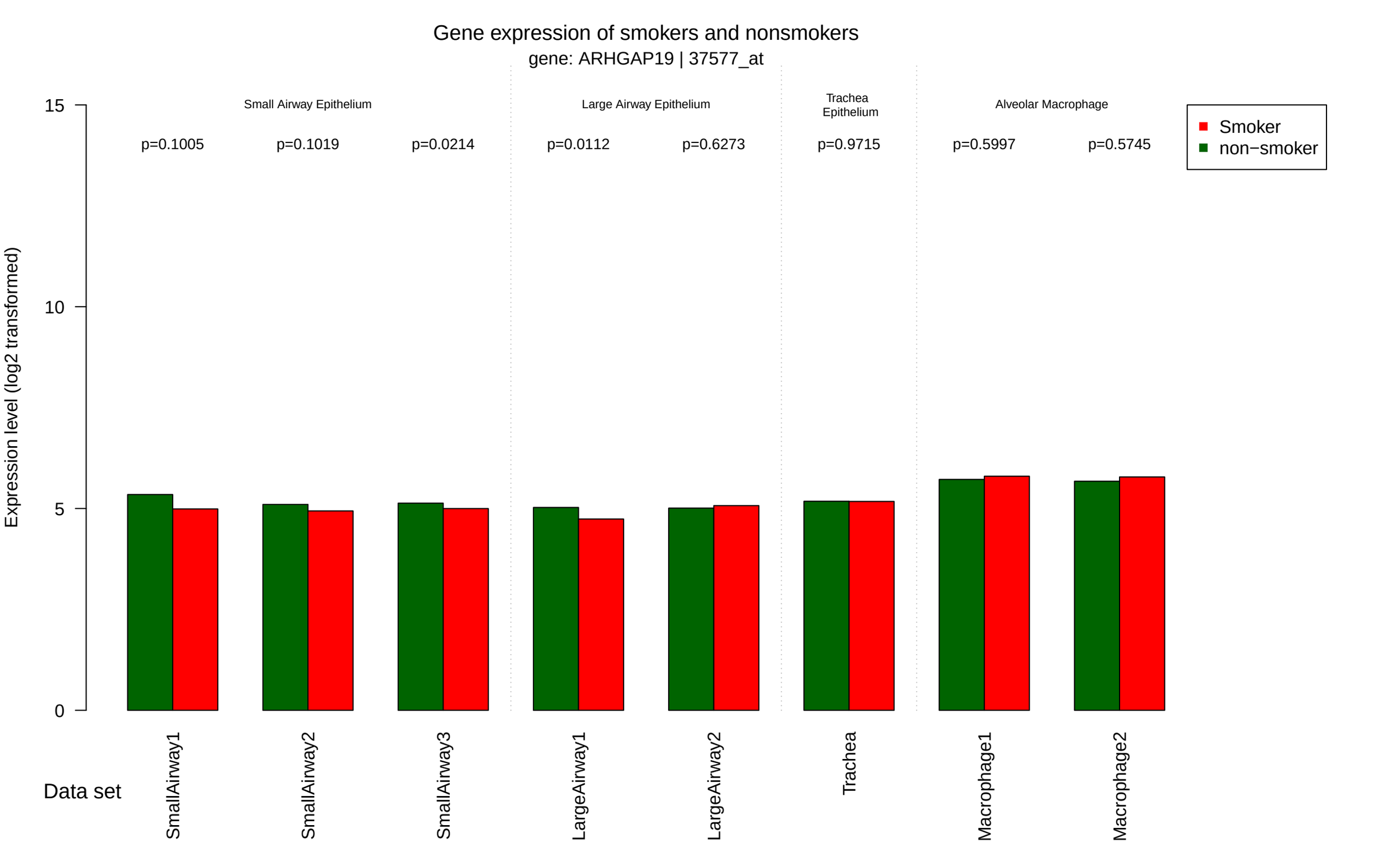
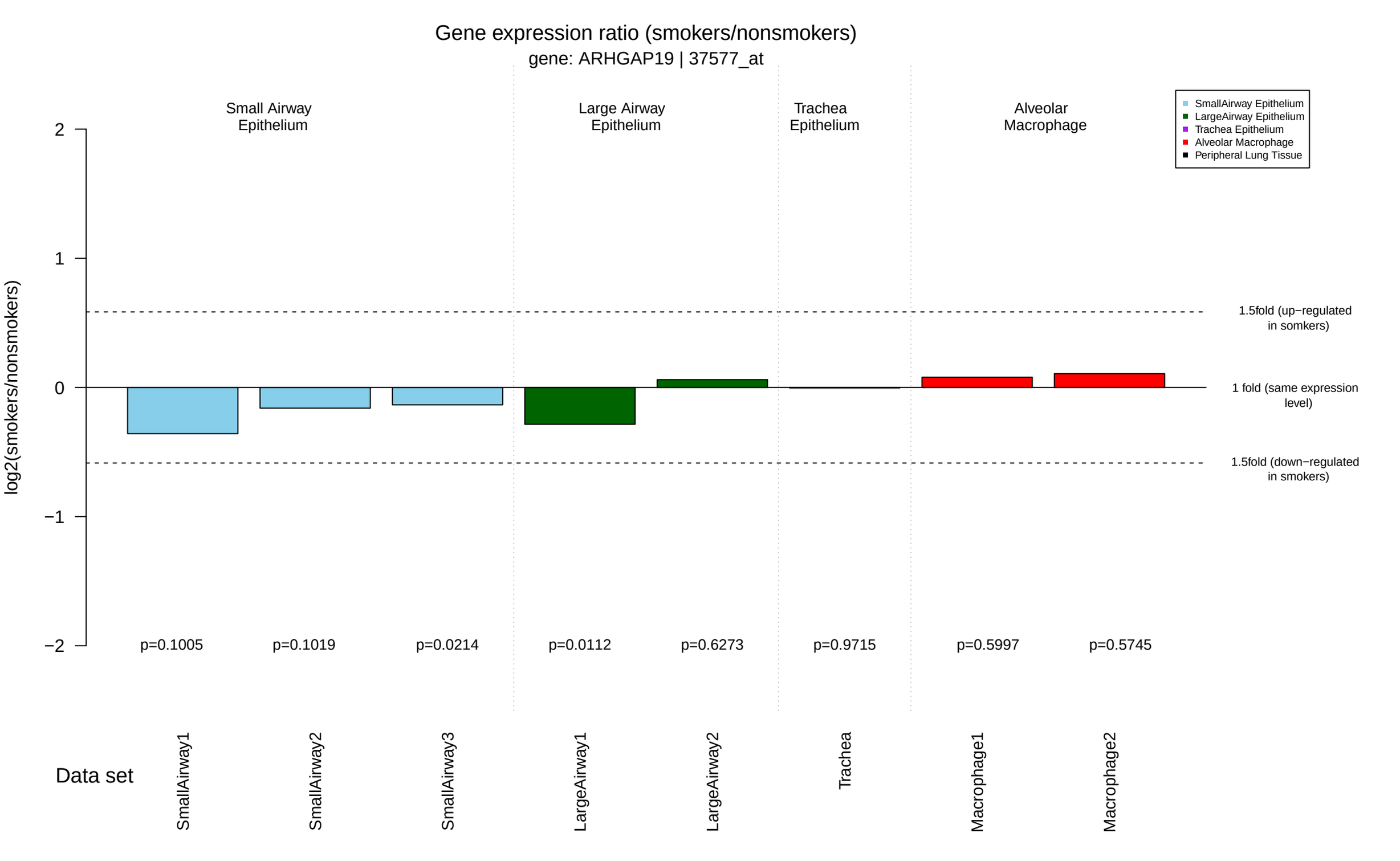
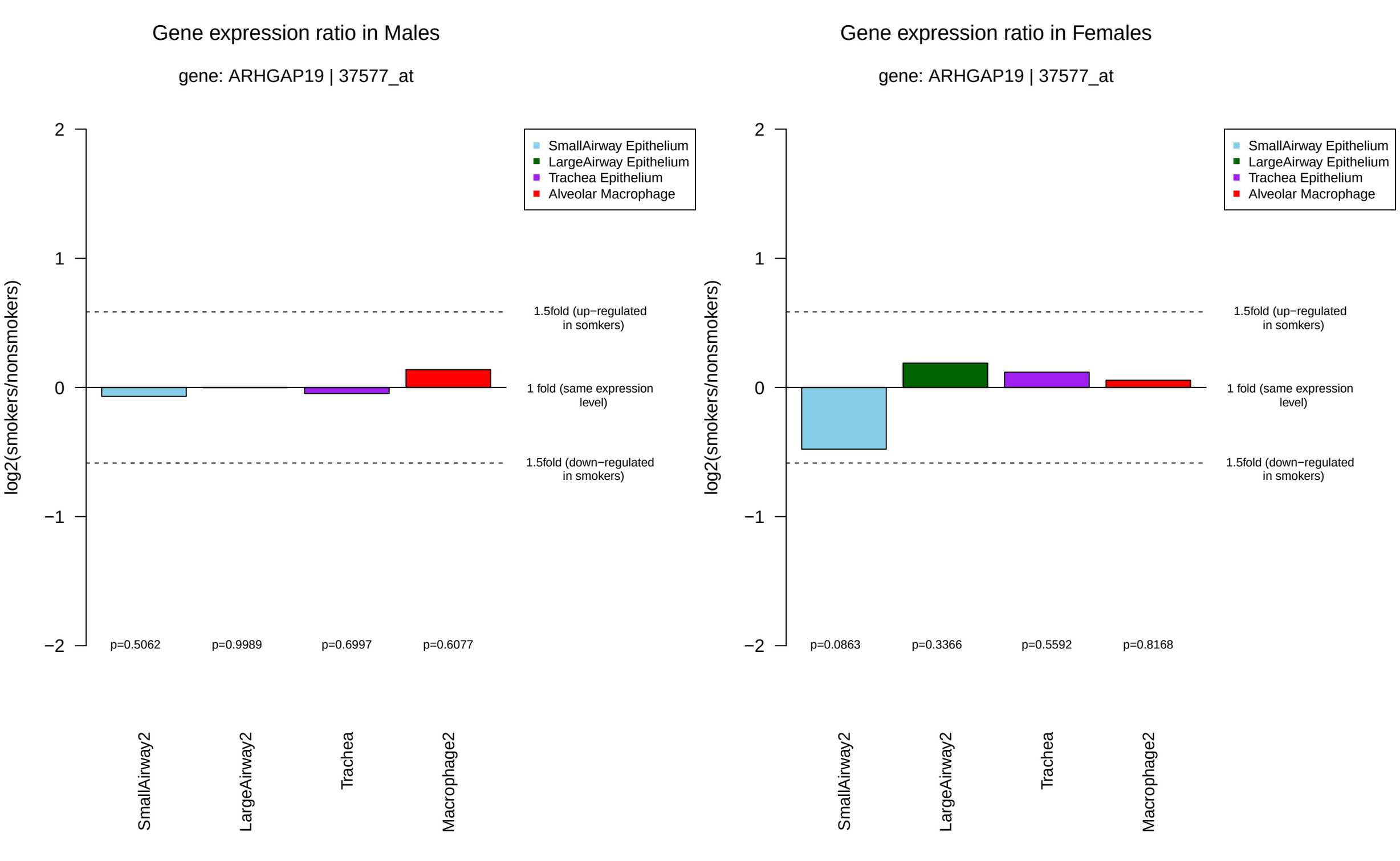
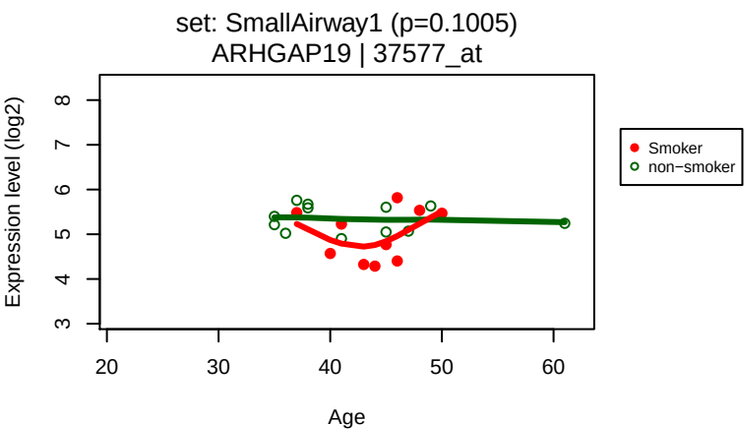
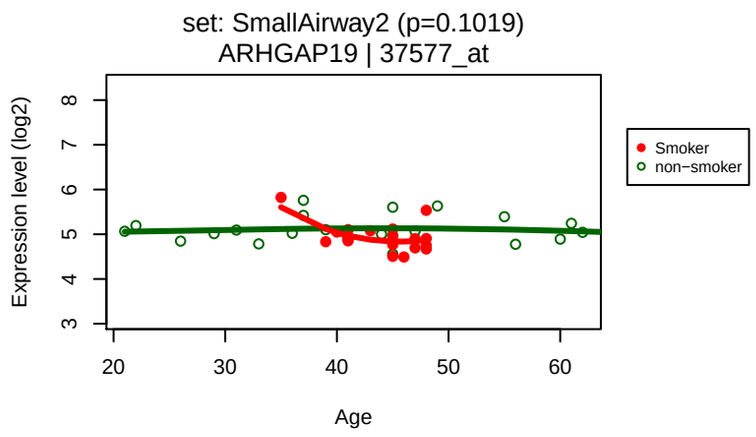
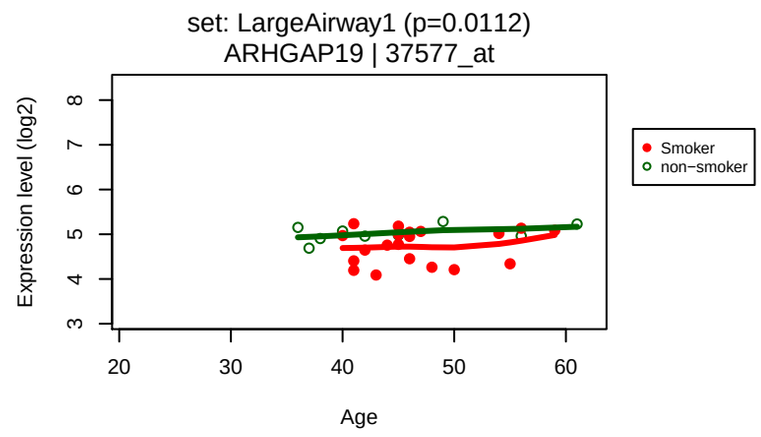

Gene expression of smokers and nonsmokers
gene: ARHGAP19 | 37577_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1005
p=0.1019
p=0.0214
p=0.0112
p=0.6273
p=0.9715
p=0.5997
p=0.5745
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

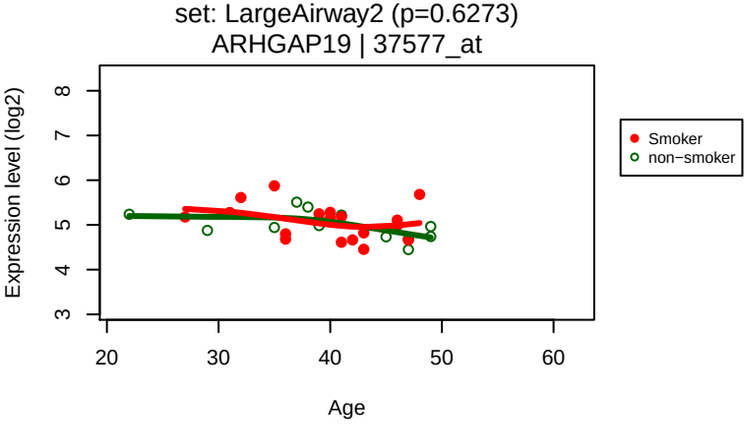

Gene expression ratio (smokers/nonsmokers)
gene: ARHGAP19 | 37577_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1005
p=0.1019
p=0.0214
p=0.0112
p=0.6273
p=0.9715
p=0.5997
p=0.5745
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

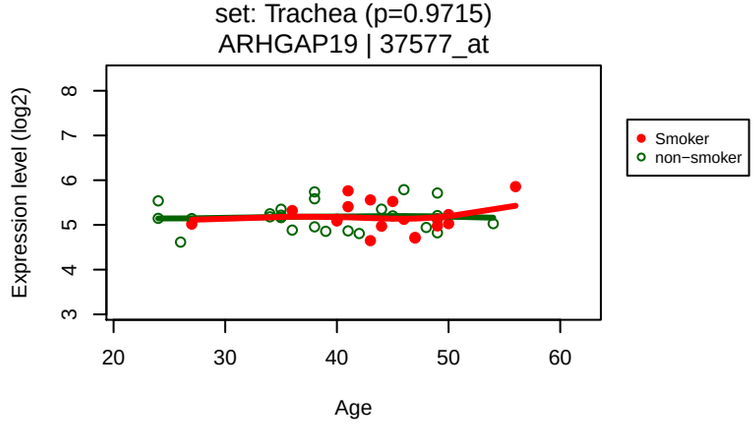

Gene expression ratio in Males
gene: ARHGAP19 | 37577_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5062
p=0.9989
p=0.6997
p=0.6077
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARHGAP19 | 37577_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0863
p=0.3366
p=0.5592
p=0.8168
SmallAirway2
LargeAirway2
Trachea
Macrophage2

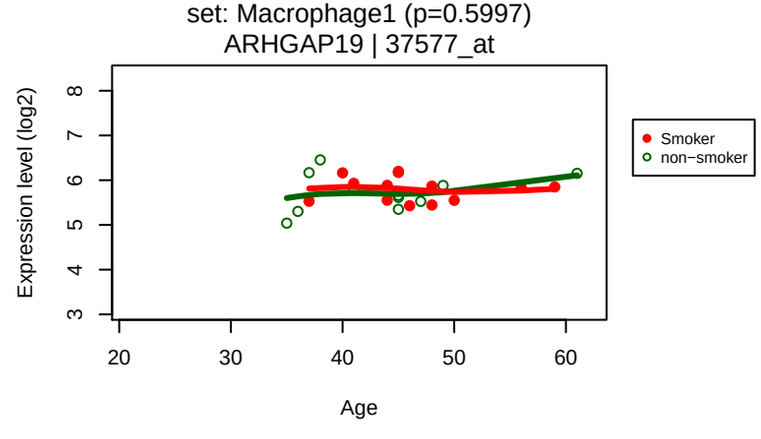

set: SmallAirway1 (p=0.1005)
ARHGAP19 | 37577_at
Expression level (log2)
Age
Smoker
non−smoker

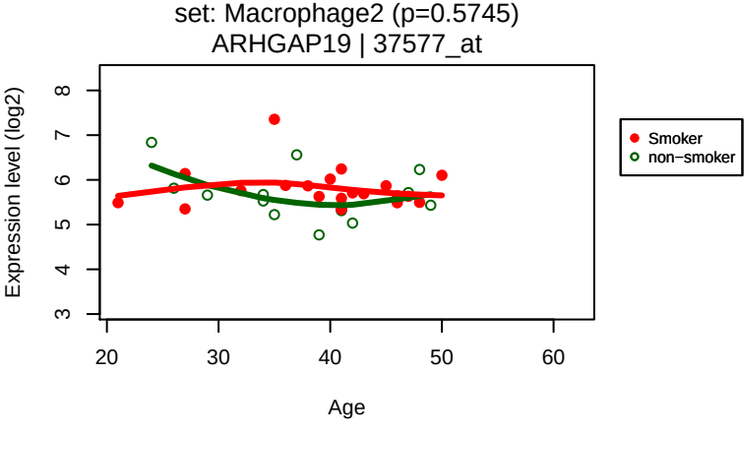

set: SmallAirway2 (p=0.1019)
ARHGAP19 | 37577_at
Expression level (log2)
Age
Smoker
non−smoker

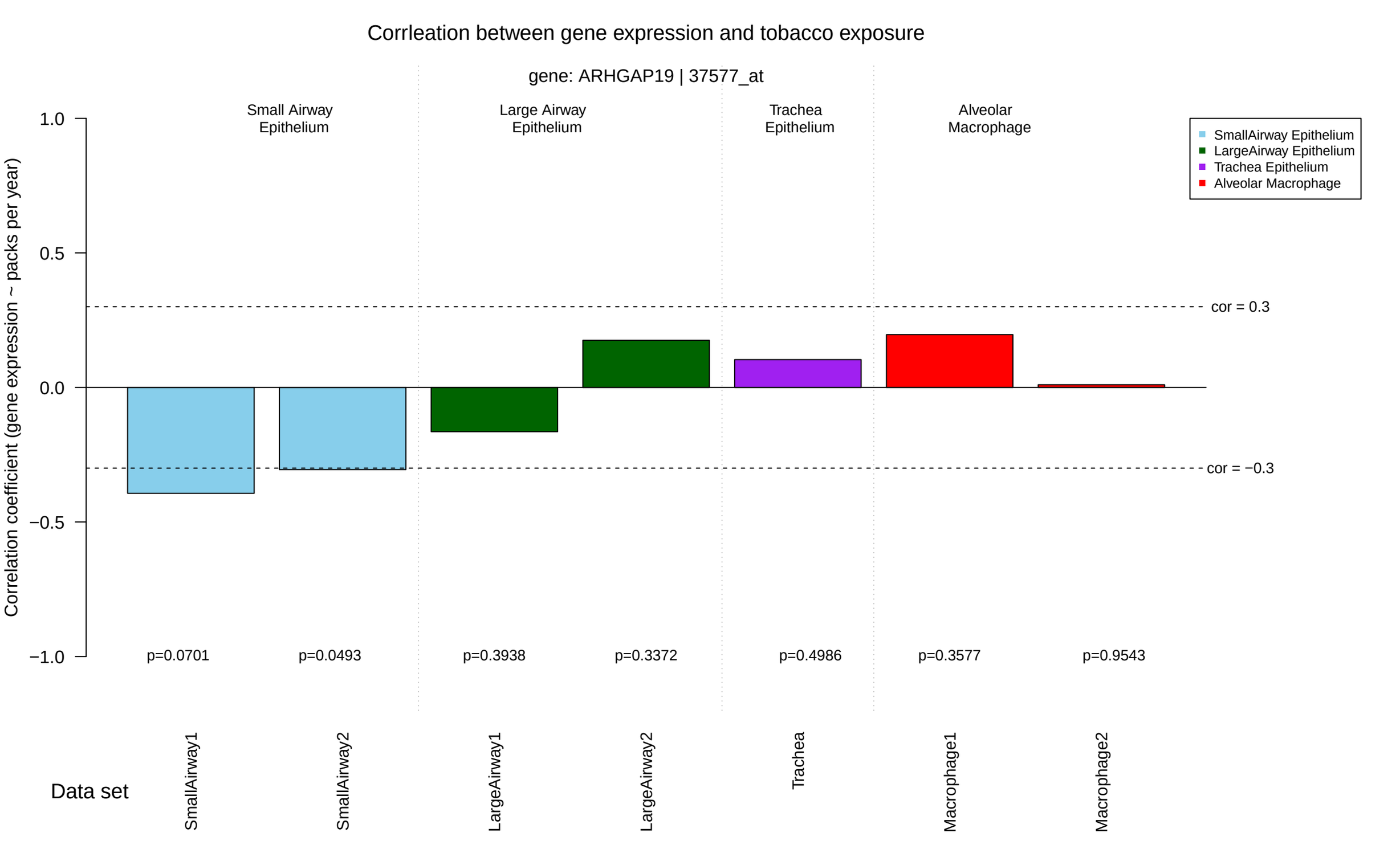

set: LargeAirway1 (p=0.0112)
ARHGAP19 | 37577_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6273)
ARHGAP19 | 37577_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9715)
ARHGAP19 | 37577_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5997)
ARHGAP19 | 37577_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5745)
ARHGAP19 | 37577_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARHGAP19 | 37577_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0701
p=0.0493
p=0.3938
p=0.3372
p=0.4986
p=0.3577
p=0.9543
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2